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Major Project
On

Context-Aware Brand Name Classification using DistilBERT-Enhanced NLP Features from corporate Websites Metadata

(Submitted in partial fulfilment of the requirements for the award of Degree)

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In

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DEPARTMENT OF INFORMATION TECHNOLOGY

CERTIFICATE

This is to certify that the project entitled "**Context-Aware Brand Name Classification using DistilBERT-Enhanced NLP Features from corporate Websites Metadata**" being submitted by **G.Prachetana (227R1A1289), G.Praneeth Reddy (227R1A1288) & D.Varun (227R1A1283)** in partial fulfilment of the requirements for the award of the degree of B.Tech in Information Technology to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

Brand identity is a critical component of business success, and generating innovative, contextually relevant, and market-ready brand names is a challenging task. Traditional approaches rely on human creativity, brainstorming sessions, or heuristic rules, which are time-consuming, subjective, and often fail to capture semantic relationships or market trends. The objective of this project is to develop an AI-powered platform that leverages natural language processing (NLP) and machine learning to automate and enhance the generation and evaluation of brand names. The proposed system preprocesses textual datasets by performing cleaning, tokenization, lemmatization, and stopword removal, followed by feature extraction using transformer-based embeddings such as DistilRoBERTa. These embeddings capture nuanced semantic patterns and contextual relevance of words and phrases. The system employs robust classifiers including Logistic Regression, Random Forest, and SVM, trained on balanced datasets using techniques like SMOTE to handle class imbalances. The platform supports both single-instance and batch predictions, providing users with predicted brand categories, suitability scores, and evaluation metrics through an intuitive web interface. This approach improves accuracy, scalability, and automation in brand name generation while reducing human effort, subjectivity, and time. By combining advanced NLP techniques with machine learning pipelines, the system demonstrates a practical and data-driven solution for businesses to innovate, evaluate, and select impactful brand names efficiently in dynamic market environments. The system preprocesses textual data through steps such as tokenization, stopword removal, and lemmatization to improve data quality. It then extracts semantic features using transformer embeddings from DistilRoBERTa, which helps capture contextual relationships within the text. Machine learning algorithms such as Logistic Regression, Random Forest, and Support Vector Machine are applied to classify and evaluate potential brand names effectively. Additionally, techniques like SMOTE are used to handle class imbalance and improve model accuracy.

The developed model is integrated with a web interface using Flask, allowing users to upload data, analyze results, and obtain predictions in real time. The proposed system reduces manual effort, enhances creativity in brand name selection, and provides an intelligent and efficient solution for businesses seeking innovative brand identities

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1.INTRODUCTION

1. INTRODUCTION

In the modern digital world, creating a unique and meaningful brand name is essential for building a strong brand identity. Traditional brand name generation methods rely mainly on manual brainstorming, which can be time-consuming and less effective. This project proposes an intelligent system that uses Artificial Intelligence and Natural Language Processing to assist in generating and evaluating brand names. The system analyzes textual data, extracts meaningful features using DistilRoBERTa embeddings, and applies machine learning algorithms to classify and predict suitable brand names. The developed model is integrated with a web interface using Flask to provide an easy and efficient platform for users to analyze and generate brand names.

This project focuses on developing an intelligent brand name generation and evaluation system using NLP and machine learning techniques. The system performs text preprocessing, extracts meaningful features using transformer-based embeddings, and applies machine learning algorithms to analyze and classify brand names.

1.1 PROJECT PURPOSE

The purpose of this project is to develop an intelligent system that helps generate and evaluate brand names using Artificial Intelligence and Natural Language Processing. It aims to simplify the brand naming process by analyzing textual data and providing meaningful suggestions.

The system uses transformer embeddings from DistilRoBERTa and machine learning algorithms to classify and analyze brand names effectively. The model is integrated with a web application using Flask, allowing users to easily interact with the system and obtain predictions.

1.2 PROJECT FEATURES

The proposed Brand Name Generation and Evaluation System is designed to assist businesses and startups in creating effective brand names using advanced technologies. The system integrates techniques from Artificial Intelligence and Natural Language Processing to analyze textual data and generate meaningful insights. One of the key features of the project is the use of modern machine learning models that can understand language patterns and evaluate brand names based on contextual meaning. This helps improve the creativity, relevance, and uniqueness of generated brand names.

Another important feature of the system is the text preprocessing module. Before applying machine learning techniques, the input textual data is cleaned and prepared through several preprocessing steps. These steps include tokenization, stopword removal, punctuation removal, and lemmatization. These processes help remove unnecessary information from the text and convert it into a structured format that can be effectively analyzed by machine learning models. Proper preprocessing improves the accuracy and performance of the model by ensuring that only meaningful textual features are used for analysis.

The system also performs feature extraction using transformer-based embeddings. In this project, a transformer model called DistilRoBERTa is used to extract deep semantic features from textual data. Transformer models are capable of understanding contextual relationships between words and capturing the meaning of sentences more effectively than traditional methods. By using these embeddings, the system can analyze brand names based on their semantic representation, which helps in producing more accurate predictions and classifications.

Another significant feature of the system is the use of multiple machine learning algorithms for classification and evaluation. Algorithms such as Logistic Regression, Random Forest, and Support Vector Machine are used to train and test the model. These algorithms analyze the extracted features and classify brand names based on learned patterns from the dataset. Using multiple algorithms allows the system to compare their performance and select the most suitable model for accurate prediction.

The project also includes handling of class imbalance in the dataset using advanced techniques. In many machine learning datasets, some classes may contain more data samples than others, which can affect the training process. To overcome this issue, the system applies

data balancing techniques such as SMOTE (Synthetic Minority Oversampling Technique).

This technique generates additional samples for minority classes, helping to create a balanced dataset and improving the reliability of the trained model.

Another important feature of the project is Exploratory Data Analysis (EDA). EDA helps in understanding the characteristics and patterns present in the dataset before model training. Various visualization techniques such as word frequency analysis, unigram and bigram analysis, and word cloud generation are used to explore the textual data. These visualizations help identify commonly used words, patterns, and relationships within the dataset, which further assists in improving the model design and performance.

The system also provides model evaluation and performance analysis. After training the machine learning models, different evaluation metrics are used to measure their performance. Metrics such as accuracy, precision, recall, and F1-score are used to determine how well the model is performing. These evaluation techniques help identify the strengths and weaknesses of the model and ensure that the final system produces reliable predictions.

Another key feature of the project is the web-based user interface. The entire system is integrated into a web application using the Flask framework. This web interface allows users to interact with the system easily. Users can upload datasets, input brand names, and view prediction results through a simple and user-friendly interface. The web application makes the system accessible and practical for real-world use.

The project also supports real-time prediction and analysis. Once the model is trained and integrated with the web interface, users can input brand names and instantly obtain evaluation results. The system analyzes the input text using the trained model and provides predictions regarding the effectiveness or classification of the brand name. This feature helps businesses quickly test and evaluate potential brand names before finalizing them.

Overall, the project provides a comprehensive platform that combines advanced language models, machine learning algorithms, and interactive web technologies to assist in brand name generation and evaluation. By automating the brand naming process and providing data-driven insights, the system reduces manual effort and helps businesses create innovative and impactful brand identities.

2.LITERATURE SURVEY

2. LITERATURE SURVEY

Marco Lemos, et al. [7] explored techniques and datasets used to train AI models as well as strategies for leveraging objective data to validate the brandability of generated names. Emotional and semantic aspects of brand names, which are often overlooked in traditional approaches, are discussed as well. A list with more than 75 pivotal datasets is presented. As a result, this review provided an understanding of the potential applications of AI, NLP, and affective computing in brand name generation and evaluation, offering valuable insights for entrepreneurs and researchers alike.

Anuj Kumar, et al. [8] underscored the importance of responsible AI usage, transparency, and continuous adaptation to changing consumer behaviors for maintaining trust and ethical standards. This contribution to the existing literature combines real-world examples, practical insights, and a mixed-methods approach, offering a unique perspective on how AI, particularly Chat GPT, can reshape customer engagement, brand communication, and creativity in both academic and industrial contexts. The article provides a comprehensive examination of AI tools' practical utility, bridging theory and application for a nuanced understanding in the field of brand marketing.

Musaiqer, et al. [9] examined the existing literature on the use of AI in brand building, including its potential to improve customer engagement, personalize marketing strategies, and optimize brand messaging. they also highlighted the challenges associated with AI implementation, such as the need for high-quality data, potential biases in algorithms, and the risk of losing the human touch in brand interactions.

Maria DSouza Deryl, et al. [10] reviewed the theoretical underpinnings of AI-driven branding from the extant literature. The thematic analysis of theoretical premises instrumentalises developing a novel integrated theoretical framework. The intricate analysis of the constellation of theories and the simultaneous amalgamation of concepts allows for a unified viewpoint. The integrated theoretical framework offers an in-depth overview of the drivers, mediators, moderators, and consequences of AI-driven branding.

Sabeh et al. [11] presented OpenBrand, a model architecture designed for extracting brand values from unstructured product titles. By incorporating both word and character-level embeddings, OpenBrand outperforms existing state-of-the-art models across various categories. The model demonstrates strong generalization capabilities, particularly in zero-shot extraction scenarios, and performs well even with compound brand values. Further, the problem of unseen words in brand names, which can arise due to sub-branding, brand fragmentation, or emerging businesses, is addressed.

Harry Ph. Sophocleous et, al. [12] explored the intersection of these technologies with entrepreneurship,

highlighting how they enhanced decision-making, customer insights, and operational efficiency. While the benefits are substantial, integrating these technologies presents challenges related to data ethics, privacy, algorithmic bias, sustainability, accessibility, and implementation complexity. Through analysis of current literature, critical perspectives, and illustrative case studies, this entry emphasises the need for strategic alignment, ethical considerations, and adaptive organisational cultures.

Mendes et al. [13] their research evaluated the performance of multilingual DistilBERT and XLM-RoBERTa models in predicting affective ratings from textual content. The assessment took into account how model size affects the accuracy of predictions; in order to determine how model complexity influences the precision of valence and arousal predictions, several transformer model sizes were tested.

Michael Gerlich et, al. [14] aimed to understand whether AI systems can replace human influencers in shaping purchasing decisions and, if so, in which sectors. A mixed-methods approach was employed, involving a quantitative survey of 478 participants with prior experience using both AI tools and interacting with social media influencers, complemented by 15 semi-structured interviews.

Talha Ahmed Khan et, al. [16] examined case examples to emphasize successful implementations, providing insights into the real advantages and knowledge gained from these efforts. Moreover, it tackles the inherent difficulties, like the complexities of integration, concerns about data privacy, and ethical considerations, that come with the use of AI in industrial operations.

Jiang et al. [17] introduced a hybrid Word2vec network layer to extract features from the text, mining the word context semantics to obtain the text vector, which was then used as the input of the LSTM layer network also created two types of benchmark models by combining text vectorization with various ML techniques such as, random forest (RF), logistic regression (LR), support vector machine (SVM), k-nearest neighbors (KNN), and hash trick (HT). Their study found that the Word2vec–LSTM hybrid model outperformed the benchmark models in terms of prediction results.

Wang et al. [18] presented a comprehensive approach for implementing an algorithm that checks for duplicated text based on semantic natural language analysis. Their models integrated advanced technologies such as Word2vec and latent Dirichlet allocation (LDA).

Liu et al. [19] proposed a paraphrase generation model which is an extension of the traditional S2S model incorporating an attention mechanism and utilizing topic words as prior knowledge to improve the quality of generated paraphrases. In order to improve the semantic reference for rewriting, the model includes a way to add subject terms, which are extracted using LDA.

Sadhuram et al. [20]. proposed an implementation phase involves using AI and NLP for better understanding and answering of questions. For example, their QA system includes question processing, passage retrieval, sentence retrieval, answer processing, and text summarization. The system was tested on over 422 articles of the SQUAD dataset and approximately 87,599 cross-domain questions.

2.1 REVIEW OF RELATED WORK

The process of creating effective brand names has become increasingly important in modern business and marketing environments. A strong brand name helps organizations build identity, improve customer recognition, and establish market presence. Traditionally, brand name generation relied heavily on human creativity, brainstorming techniques, and linguistic analysis. However, these manual approaches can be time-consuming and may not always produce innovative or data-driven results. With the advancement of Artificial Intelligence and Natural Language Processing (NLP), researchers have started exploring automated systems that can generate and evaluate brand names using machine learning techniques. Recent studies focus on text processing, semantic analysis, and classification models to analyze linguistic patterns and generate meaningful brand name suggestions.

2.1.1 Natural Language Processing in Text Analysis

Natural Language Processing plays a significant role in analyzing and understanding textual data. NLP techniques such as tokenization, stopword removal, stemming, and lemmatization are commonly used to preprocess text before applying machine learning algorithms. These preprocessing methods help transform raw textual data into structured formats that can be analyzed effectively. NLP has been widely used in various applications such as sentiment analysis, text classification, and name generation systems.

2.1.2 Machine Learning for Text Classification

Machine learning algorithms have been widely used for analyzing and classifying textual data. Models such as Logistic Regression, Random Forest, and Support Vector Machines are commonly applied to identify patterns and relationships within text datasets. These algorithms help classify words and phrases based on their semantic meaning and contextual relevance. By training these models on labeled datasets, systems can automatically predict or evaluate the quality and suitability of brand names.

2.1.3 Transformer Models for Semantic Understanding

Recent advancements in deep learning have introduced transformer-based language models that significantly improve text understanding. Models such as **DistilRoBERTa** are capable of generating contextual embeddings that capture deeper semantic relationships between words. These models provide more accurate feature representations compared to traditional techniques. Using transformer embeddings allows systems to analyze brand names more effectively and improves the overall prediction performance of machine learning models.

2.1.4 Data Balancing and Model Performance Improvement

In many real-world datasets, class imbalance can affect the performance of machine learning models. Techniques such as Synthetic Minority Over-sampling Technique (SMOTE) are used to balance datasets by generating synthetic samples for minority classes. This approach improves the training process and helps models achieve better accuracy and reliability when classifying textual data.

2.1.5 Web-Based AI Systems for User Interaction

Modern AI systems are often integrated with web frameworks to make them accessible and interactive for users. Frameworks like Flask allow developers to deploy machine learning models through web interfaces where users can upload data, analyze results, and obtain predictions in real time. Such integration makes AI-based brand name evaluation systems more practical and user-friendly.

2.2 DEFINITION OF PROBLEM STATEMENT

Classifying companies into their correct GICS Industry Groups based solely on unstructured textual descriptions is complex due to inconsistent wording, varied terminology, and semantic ambiguity across industries. Traditional NLP techniques often fail to understand contextual meaning, causing inaccurate classification in multi-class setups. The problem addressed in this research is to develop an automated, context-aware, and scalable system that uses

DistilRoBERTa embeddings and machine learning classifiers to accurately predict GICS Industry Groups from raw text while overcoming issues such as class imbalance, noise, and inconsistent language structures.

With the advancement of technologies such as Artificial Intelligence and Natural Language Processing, it is possible to design intelligent systems that can understand textual data and identify meaningful patterns in language. These technologies allow computers to analyze the semantic relationships between words and provide data-driven insights. By using advanced language models like DistilRoBERTa, textual features can be extracted more effectively, enabling accurate analysis and classification of brand names.

Therefore, the main problem addressed in this project is the development of an intelligent system that can assist in generating and evaluating brand names automatically. The system aims to reduce the limitations of manual methods by applying machine learning techniques to analyze textual data and provide meaningful predictions. The final solution is implemented through a web-based interface using Flask, allowing users to easily interact with the system and obtain quick results. This approach helps simplify the brand naming process and supports businesses in selecting creative, relevant, and impactful brand names.

2.3 EXISTING SYSTEM

Before the integration of Artificial Intelligence and NLP, brand name generation and evaluation were conducted using conventional, human-driven techniques. These traditional systems relied on creative brainstorming, linguistic analysis, and keyword-based suggestions rather than computational intelligence. Marketing teams manually generated, evaluated, and selected names through group discussions and linguistic adjustments. Some organizations adopted simple software tools to generate word combinations or apply phonetic rules, but these systems lacked semantic understanding, creativity, and scalability. As a result, the process was time-consuming, subjective, and limited in producing innovative and globally appealing brand names.

Limitations of Existing System

1. Highly time-consuming and depends on multiple brainstorming sessions.
2. Results are subjective and vary based on team creativity and individual bias.
3. Scalability is limited, as human-driven ideation cannot handle large datasets.
4. Difficult to maintain consistency across different branding projects.
5. Lacks data-driven insights, making it hard to measure linguistic or emotional impact.

2.4 PROPOSED SYSTEM

The proposed system employs a hybrid Natural Language Processing (NLP) and Machine Learning pipeline to classify industry groups using text data. The process begins with dataset loading and preprocessing, where raw text is cleaned, tokenized, lemmatized, and stopwords are removed using NLTK. Processed data is saved for reusability, and label encoding is applied for categorical targets. Exploratory Data Analysis (EDA) visualizations, including word clouds, POS tagging, and bigram analysis, help in understanding text structure and patterns. Semantic embeddings are extracted using the DistilRoBERTa transformer model to capture contextual relationships in textual data. The extracted embeddings are balanced using SMOTE to handle class imbalance effectively. Multiple Machine Learning algorithms, namely Logistic Regression, Random Forest, and Support Vector Classifier, are trained using these embeddings. Model performance is evaluated through metrics such as Accuracy, Precision, Recall, and F1-Score, computed by a custom Metrics Calculator module. Graphical visualizations of these metrics are generated for comparative analysis. Finally, the best-performing model is saved and used to predict industry groups for unseen test data, ensuring a robust and generalizable classification framework.

Advantages of the Proposed System

- Enhances brand evaluation accuracy through AI-based analysis.
- Reduces manual workload in brand name assessment.
- Ensures linguistic consistency and cultural adaptability across regions.
- Provides data-driven insights to guide marketing strategies.
- Minimizes risk of trademark conflicts and naming redundancies.
- Offers objective evaluation metrics for creative decision-making
- Increases naming success rate by using predictive linguistic modeling.
- Saves time and resources by automating preliminary evaluation stages.
- Strengthens collaboration between creative and analytical teams.

2.5 OBJECTIVES

- To develop an intelligent system for brand name generation and evaluation using Artificial Intelligence and Natural Language Processing.
- To preprocess and analyze textual data to extract meaningful features.
- To use transformer embeddings from DistilRoBERTa for better language understanding.
- To apply machine learning algorithms to classify and evaluate brand names.
- To integrate the system with a web interface using Flask for easy user interaction.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

2.6.1.1	Processor	:	Intel Core i3 or above
2.6.1.2	Hard disk	:	1TB
2.6.1.3	RAM	:	8GB.

2.6.2 SOFTWARE REQUIREMENTS:

2.6.2.1	Operating system	:	Windows
2.6.2.2	Language	:	Python 3.12.0
2.6.2.3	Framework	:	Flask

3. SYSTEM ARCHITECTURE & DESIGN

3. SYSTEM ARCHITECTURE & DESIGN

The system architecture of Our project includes modules for data input,text preprocessing,feature extraction,model training,and prediction.Thne input data is cleaned using NLP techniques such as tokenization and stopword removal.DistilBERTa is used for feature extraction,and machine learning algorithms like Logistic regression,Random forest,and SVM are applied for classification.The trained model is integrated with Flask web interface to provode real-time brand name evaluation.

3.1 PROJECT ARCHITECTURE

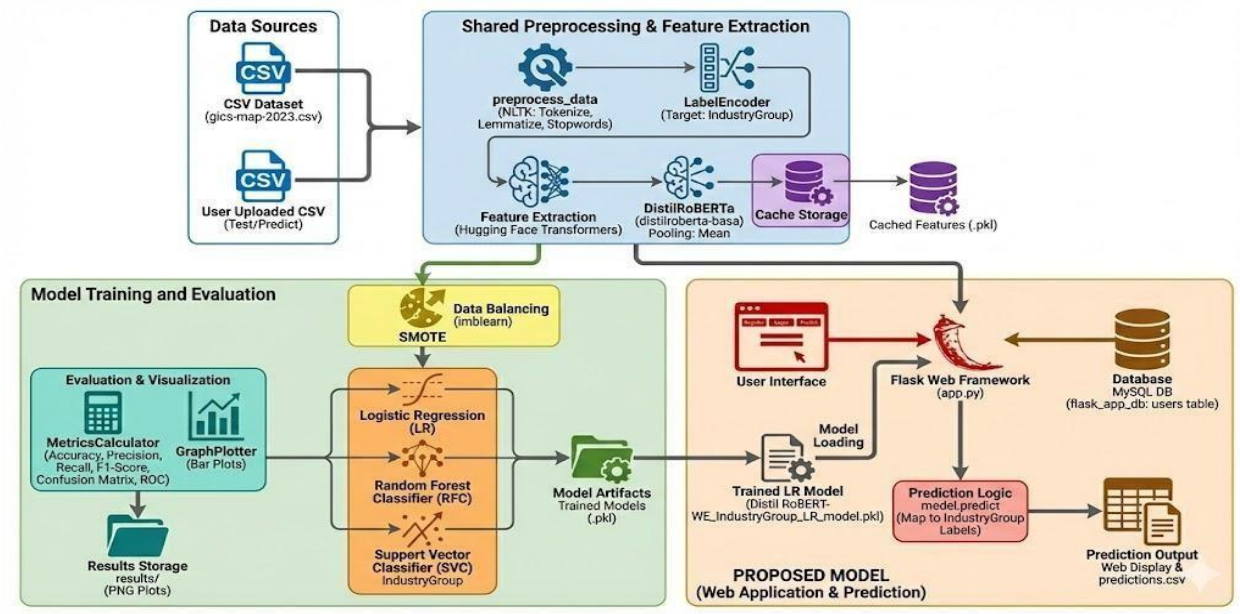


Figure 3.1: System Architecture of Context-Aware Brand Name Classification using DistilBERT Enhanced NLP Features from corporate Websites Metadata

3.2 DESCRIPTION

FlaskFramework

Flask serves as the backbone of the Brand Name Generation and Evaluation System, providing a seamless bridge between the user interface, data processing, and AI model predictions. It manages routes for exploratory data analysis, model evaluation, and prediction while ensuring a smooth user experience. Flask handles user inputs such as text or CSV uploads, processes them efficiently, and routes the data to the backend for analysis. It integrates pre-trained models, embedding features, and visualization tools to present results interactively. The framework also ensures modularity, allowing separate handling of EDA, model analysis, and predictions. Through Flask's lightweight structure, the system achieves real-time analysis without compromising speed or scalability.

UserInteractionLayer

The user interaction layer begins the workflow where users upload text or CSV files through HTML templates. Flask receives these inputs via defined routes such as **/prediction** or **/model_analysis**. This stage ensures that users can conveniently interact with the system using simple web forms.

FlaskBackendCore

This layer is the foundation of all request handling and validation operations in Flask. The route handler manages incoming GET and POST requests, directing them to appropriate processing functions. Flask performs validation checks to confirm that uploaded files or text inputs match the expected structure. It may also temporarily store validated data for later use. This ensures that only accurate and clean data passes into the preprocessing stage.

PreprocessingLayer

At this stage, the uploaded data is loaded, cleaned, and transformed into a structured form for machine learning. The preprocessing module loads cleaned datasets and precomputed embeddings generated using models such as DistilRoBERTa. Label encoding is applied to convert categorical text data into numerical form for model training and prediction. Flask

automates this process through backend functions triggered by user submissions.

3.3 UML DIAGRAMS

3.3.1 Class diagram

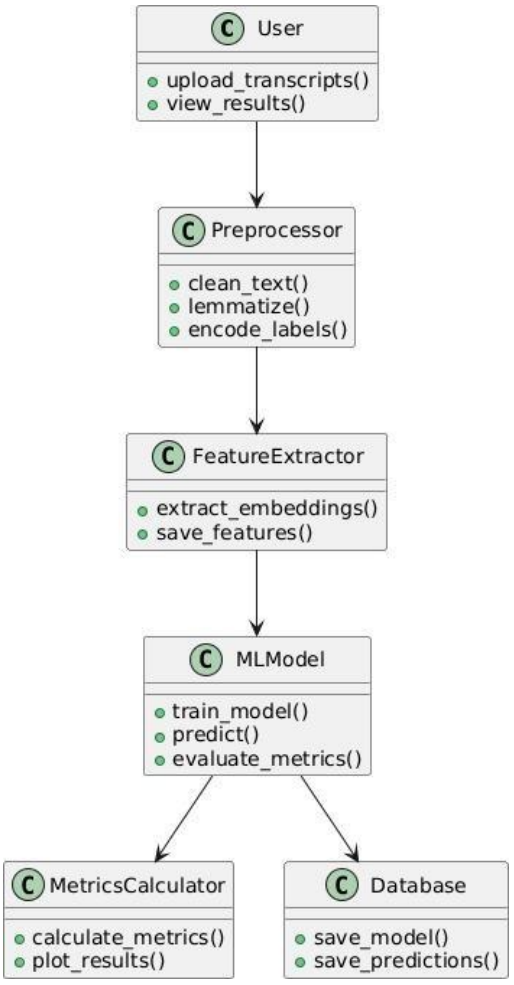


Figure 3.3.1 : Class Diagram of Context-Aware Brand Name classification using DistilBERT Enchanced NLP features from corporate website metadata

The class diagram illustrates the structural design of the proposed system, showing the key classes such as DataPreprocessor, EmbeddingGenerator, ModelTrainer, and NameEvaluator. Each class encapsulates related attributes and methods that manage text cleaning, embedding creation, model training, and evaluation.

3.3.2 Use case diagram

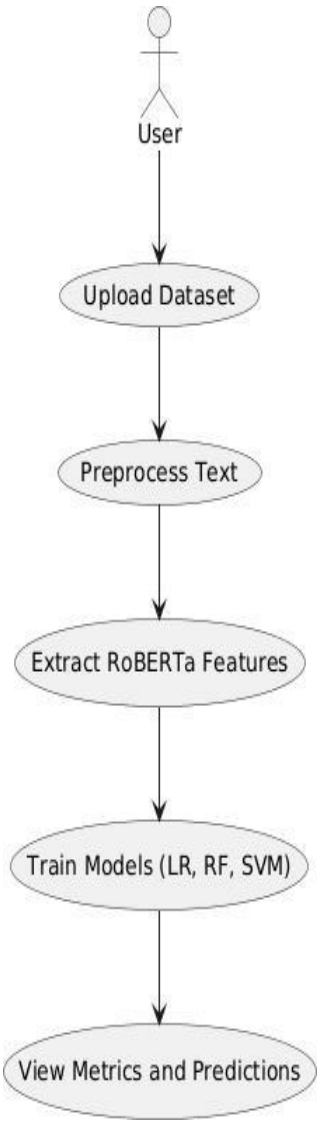


Figure 3.3.2 Use case diagram of Context-Aware Brand Name classification using DistilBERT Enchanced NLP features from corporate website metadata

The use case diagram highlights the interaction between the user and the system. It shows how users can input data, generate potential brand names, evaluate them, and visualize performance. The system's backend handles the AI and NLP operations, while the user interacts through an intuitive interface to receive optimized brand name suggestions.

3.3.3 Sequence Diagram

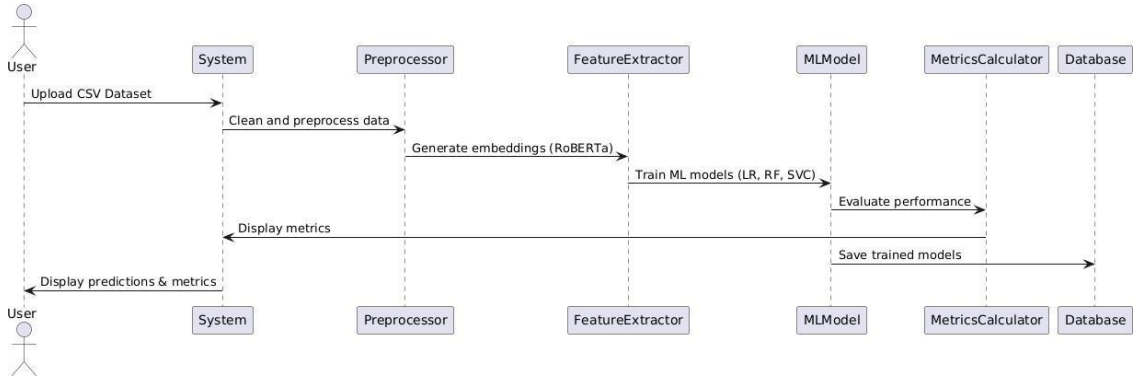


Figure 3.3.3 : Sequence diagram of Context-Aware Brand Name classification using DistilBERT Enchanced NLP features from corporate website metadata

The sequence diagram represents the dynamic flow of operations between system components. It begins with the user submitting brand data, followed by preprocessing, embedding generation, model training, and name evaluation. The sequential messages showcase how data moves and transforms across components until final results are returned to the user interface.

3.3.4 Activity Diagram

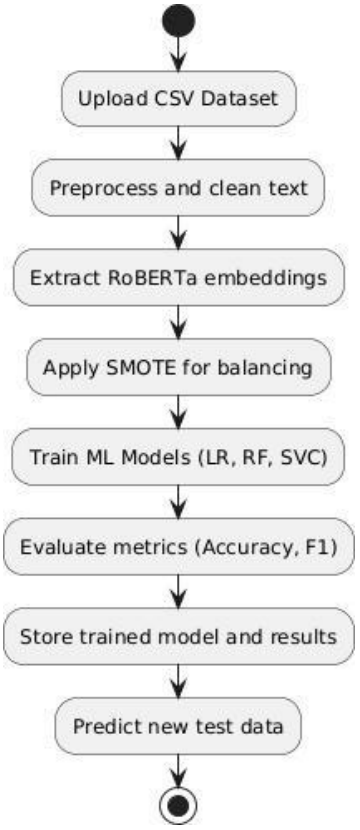


Figure 3.3.4 : Activity Diagram of Context-Aware Brand Name classification using DistilBERT Enchanced NLP features from corporate website metadata

The activity diagram visualizes the workflow of the proposed system starting from input collection, data preprocessing, embedding generation, model training, and evaluation, to final output presentation. It emphasizes the decision-making flow and concurrent activities during the brand name generation and evaluation process.

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation phase of a project involves executing the planned strategies and tasks. It requires meticulous coordination, resource allocation, and monitoring to ensure that objectives are met efficiently. Effective implementation is crucial for achieving project goals and delivering expected outcomes within the set timeline and budget constraints.

4.1 ALGORITHMS USED

Exploratory Data Analysis

Exploratory Data Analysis (EDA) helps uncover patterns, structure, and hidden insights within the dataset before model development. It allows us to understand the distribution of categories, identify noisy or inconsistent text, and detect imbalanced groups across GICS Industry classifications. Through visualizations like word clouds, frequency plots, POS-tag analysis, and document-length distributions, EDA highlights the linguistic characteristics of the dataset. This understanding is essential for designing an optimized preprocessing pipeline and ensuring transformer models receive high-quality input. Ultimately, EDA forms the foundation for building a reliable and well-generalized industry classification system.

Data Inspection & Structure Understanding: The process begins by examining dataset columns, checking missing values, unique categories, and identifying text fields. This initial scan reveals the dataset's shape, content quality, and potential preprocessing issues. It ensures the system knows exactly what type of textual and numerical data needs cleaning.

Text Cleaning & Token Analysis: Raw text is analyzed through tokenization, stopword removal, and lemmatization to observe how frequently meaningful tokens appear. By studying cleaned tokens, the system identifies redundant words, domain-specific terms, and noise that may disrupt the embedding model. This step prepares the ground for accurate feature extraction.

Word Frequency & N-gram Distribution: Word frequency plots and N-gram (bigram) visualizations highlight dominant language patterns. They help reveal repetitive terms, paired expressions, and common phrasing across industries. This insight helps determine whether the dataset carries distinct textual signatures that models can learn effectively.

Document Length & Variability Analysis: A histogram of document length shows how many words each record contains and whether the dataset is consistent or highly variable. This allows for early decisions about padding lengths, truncation strategies, and transformer token limits. It ensures the embedding model handles text efficiently without losing important context.

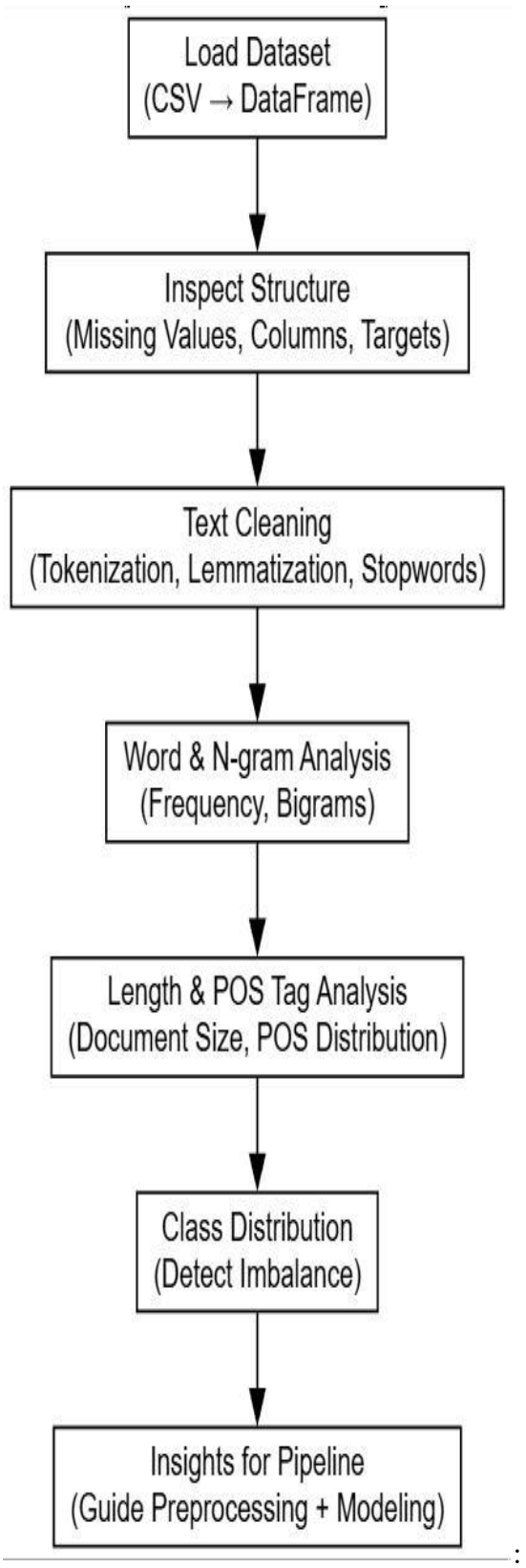


FIGURE 4.1.1 : Internal Workflow of Exploratory Data Analysis : Context-Aware Brand Name

Natural Language Processing (NLP) :

Removes unnecessary words and make sentence clean.

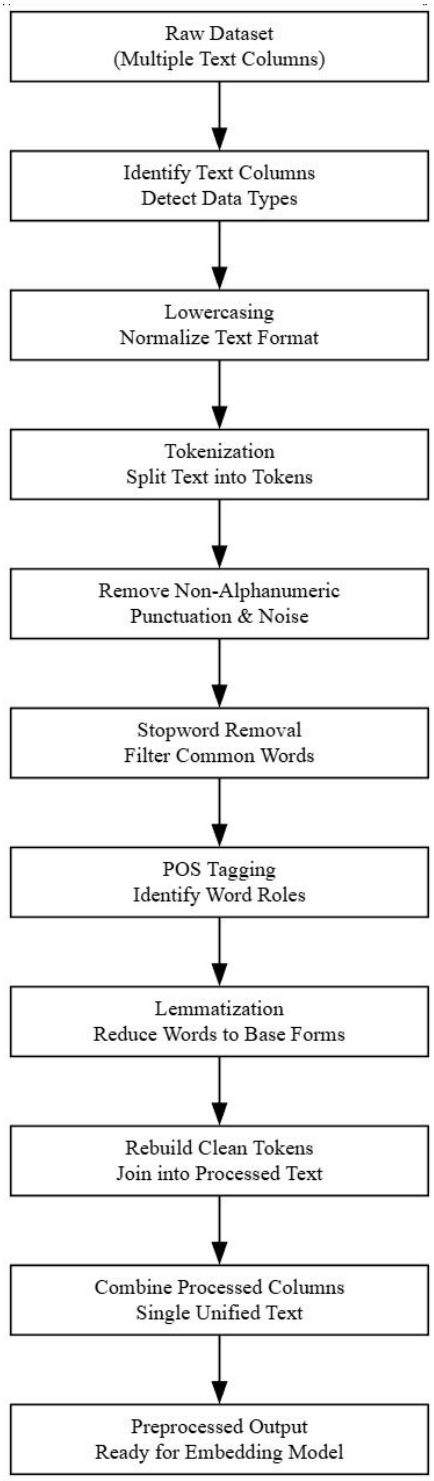


FIGURE 4.1.2 : Internal Workflow of the NLP Preprocessing **:** Context-Aware Brand Name Classification using DistilBERT-Enhanced NLP Features from corporate Websites Metadata

Transformer Models

Transformer-based language models are used to generate contextual embeddings from textual data. These embeddings capture semantic relationships between words and help improve prediction accuracy.

BERT

BERT is a transformer-based language model that understands context from both directions in a sentence.

BERT (Bidirectional Encoder Representations from Transformers) is an advanced transformer-based language model developed by Google that enables deep contextual understanding of textual data. Unlike traditional models that read sentences in a single direction, BERT reads text bidirectionally—both left-to-right and right-to-left—allowing it to understand how the meaning of each word changes depending on its surrounding context. This unique bidirectional training approach enables BERT to grasp complex linguistic structures, sarcasm, negations, and semantic nuances more accurately than conventional NLP models. BERT is pre-trained on massive datasets such as Wikipedia using unsupervised learning techniques, which helps it learn universal language representations that can be fine-tuned for specific downstream tasks like sentiment analysis, question answering, or intent classification.

RoBERTa

RoBERTa is an improved version of BERT that is trained with more data and optimized training techniques.

RoBERTa (Robustly Optimized BERT Pretraining Approach) is an improved and fine-tuned extension of BERT introduced by Facebook AI. It retains BERT's transformer architecture but optimizes its training methodology for greater efficiency and accuracy. RoBERTa removes the Next Sentence Prediction (NSP) objective and instead focuses on training with larger batch sizes, longer sequences, and more diverse data. This modification allows RoBERTa to better capture real-world language patterns, emotional expressions, and conversational tone—making it exceptionally effective for sentiment classification tasks.

RoBERTa uses Dynamic Masking during pretraining, which ensures that the masked tokens change across training epochs, leading to better generalization. Additionally, RoBERTa is trained on extensive datasets such as BookCorpus, CC-News, and OpenWebText, enabling it to understand domain-specific vocabulary and human emotional expressions more effectively than traditional BERT.

DistilRoBERTa

DistilRoBERTa is a lighter and faster version of RoBERTa that provides high-quality embeddings while reducing computational cost. It is used in this project to extract semantic features from text efficiently.

DistilRoBERTa, (Distilled Robustly Optimized BERT Approach) is a lightweight and highly efficient version of RoBERTa created through knowledge distillation. It retains nearly 95% of RoBERTa's language understanding capability while being significantly faster and smaller, making it ideal for real-time applications. Built on transformer self-attention, it captures deep semantic meaning and contextual relationships across text. Its compact architecture ensures quick inference without compromising on accuracy. Because of this balance between speed and performance, DistilRoBERTa is widely used for text encoding, classification, semantic analysis, and large-scale NLP deployments.

In this project, DistilRoBERTa is used as a feature extractor to convert each cleaned text entry into a dense semantic embedding vector. RoBERTa created through knowledge distillation. It retains nearly 95% of RoBERTa's language understanding capability while being significantly faster and smaller, making it ideal for real-time applications. Built on transformer self-attention, it captures deep semantic meaning and contextual relationships across text. Its compact architecture ensures quick inference without compromising on accuracy. Because of this balance between speed and performance, DistilRoBERTa is widely used for text encoding, classification, semantic analysis, and large-scale NLP deployments.

The code loads the pretrained "distilroberta-base" model and tokenizer from HuggingFace, tokenizes each text, applies padding/truncation, and passes it through the transformer. Instead of using the CLS token, the pipeline applies mean pooling across token embeddings to obtain a context-rich sentence representation. These embeddings are then saved and fed into machine learning classifiers (LR, RFC, SVC) for predicting the IndustryGroup labels. By using DistilRoBERTa this way, the system gains powerful contextual understanding while keeping processing fast and efficient for a live Flask application.

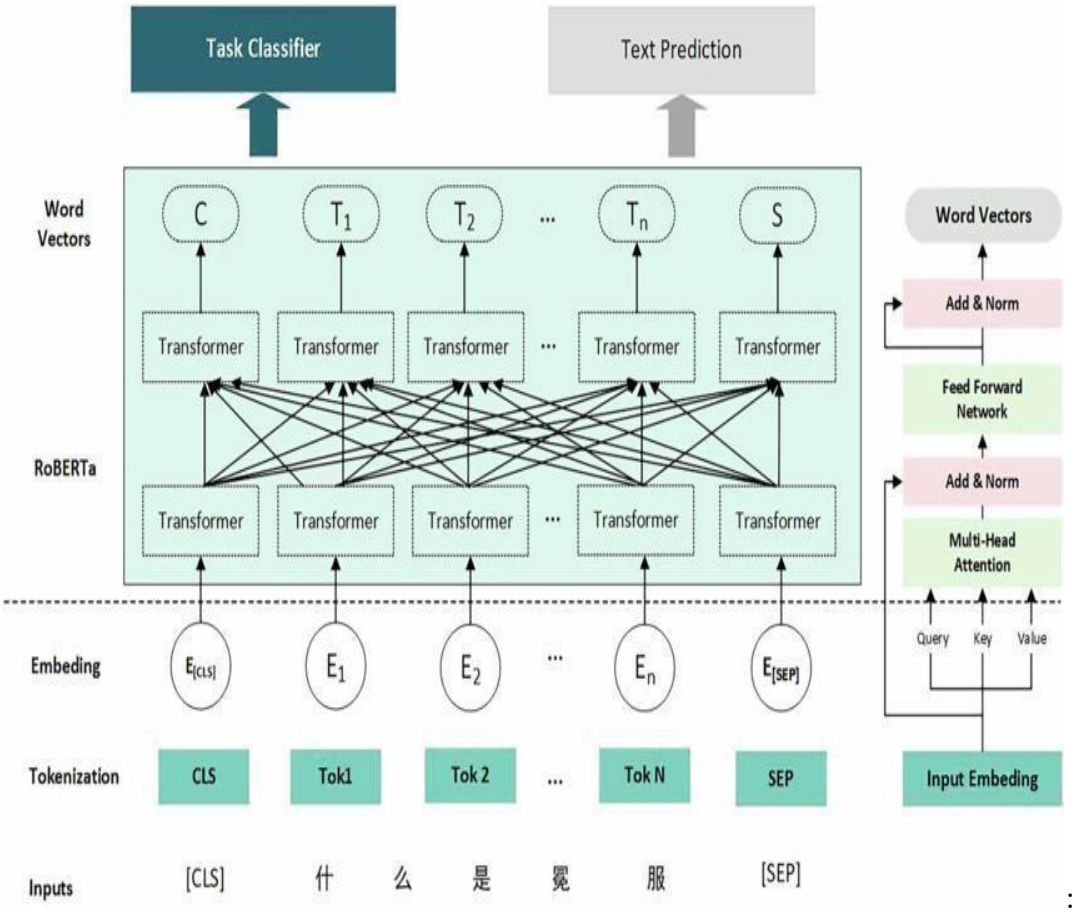


FIGURE 4.1.3 : Internal Working Operational Flow of RoBERTa : Context-Aware Brand Name Classification using DistilBERT-Enhanced NLP Features from corporate Websites Metadata

Random Forest

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction performance. Each tree is trained on a random subset of the dataset, and the final prediction is determined by majority voting among the trees.

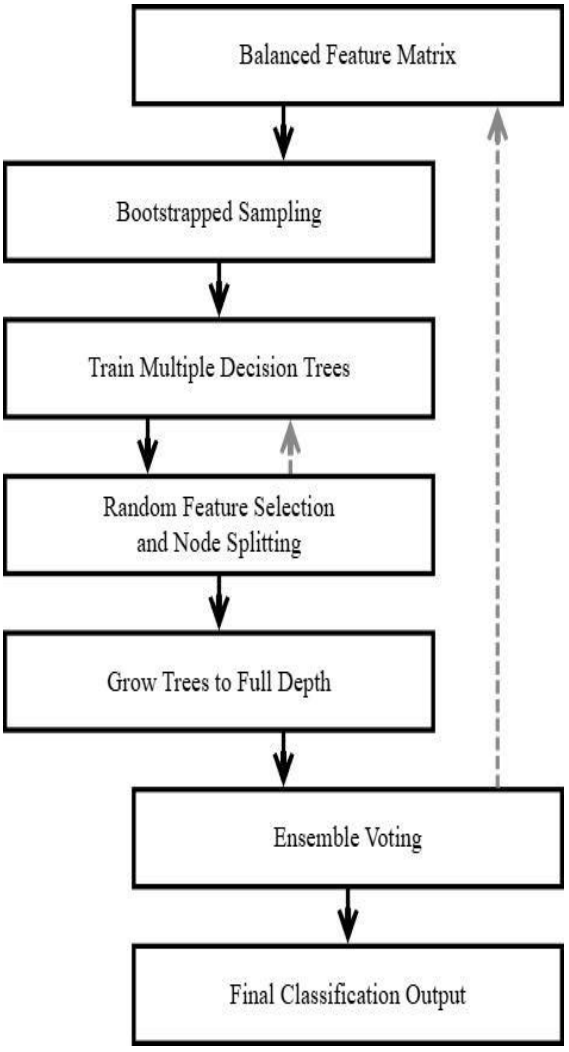


FIGURE 4.1.4 : Internal Working Flow of Random Forest : Context-Aware Brand Name classification using DistilBERT Enchanced NLP features from corporate website metadata

SMOTE Balancing

SMOTE (Synthetic Minority Over-sampling Technique) is a powerful data balancing method used to handle class imbalance in machine learning datasets. Instead of duplicating minority samples, SMOTE generates new synthetic samples by interpolating between existing minority instances. This creates more natural, diverse representations and prevents overfitting that occurs with simple replication. When working with high-dimensional embeddings like DistilRoBERTa sentence vectors SMOTE helps ensure every Industry Group category contributes equally during training. Balanced data results in more stable learning, better generalization, and fairer classification.

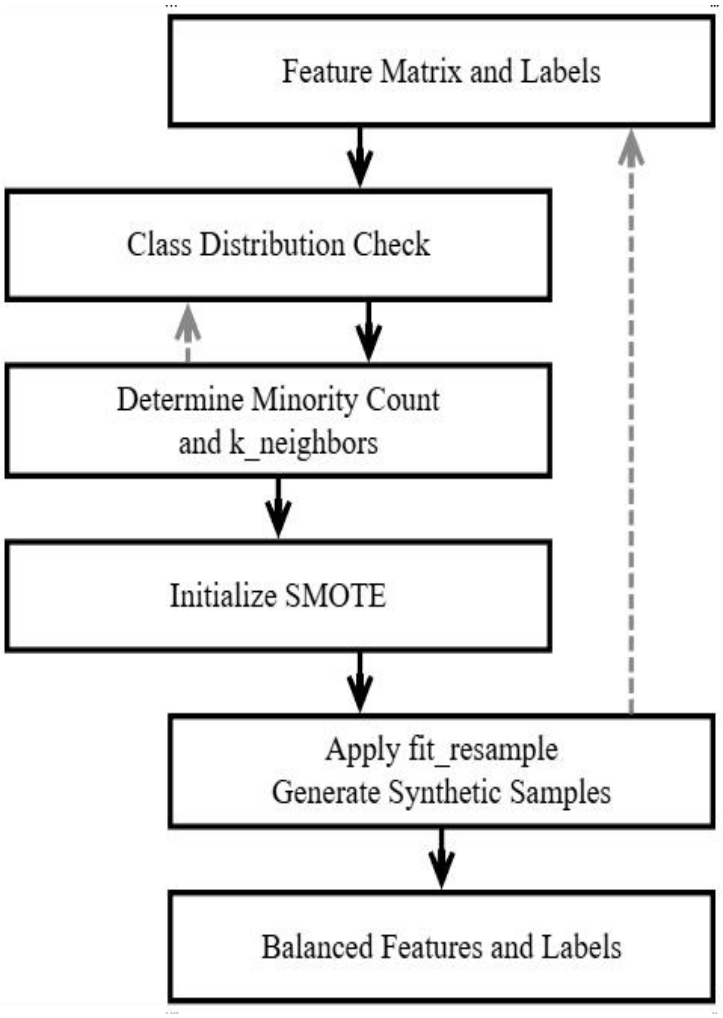


FIGURE 4.1.5 : Internal Working Flow of SMOTE Data Balancing : Context-Aware Brand Name classification using DistilBERT Enchanced NLP features from corporate website metadata

Support Vector Machine

Support Vector Machine (SVM) is a powerful supervised learning algorithm designed to find the optimal boundary that best separates different classes in a dataset. It works by identifying support vectors critical data points that define the margin between classes and maximizing this margin to improve classification accuracy. SVM can handle both linear and non-linear data using kernel functions, making it highly flexible for complex feature spaces. When applied to high-dimensional embeddings like DistilRoBERTa vectors, SVM excels at finding clean decision boundaries. Its strong generalization ability, robustness against overfitting, and effectiveness with limited samples make it a reliable choice for multi-class industry classification tasks.

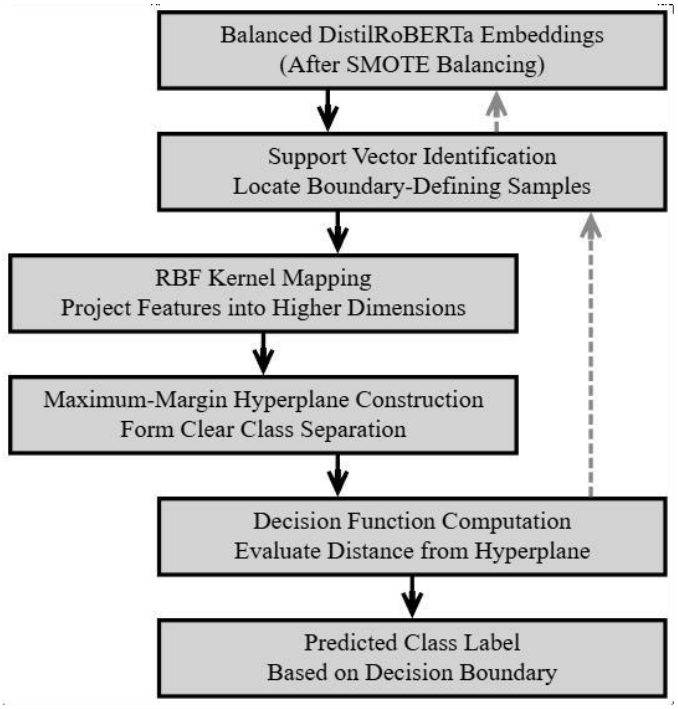
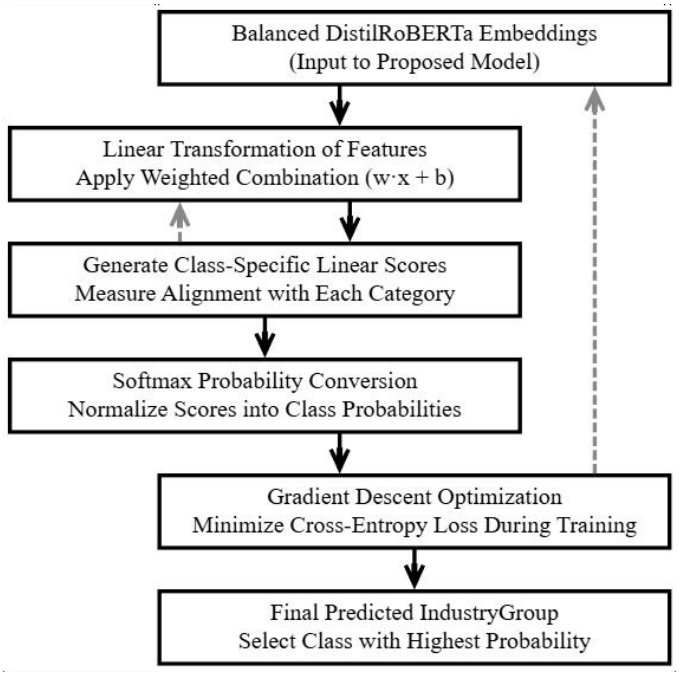


FIGURE 4.1.6 : Internal working Flow of SVM : Context-Aware Brand Name classification using DistilBERT Enchanced NLP features from corporate website metadata

Logistic Regression

Logistic Regression is a widely used statistical learning algorithm designed for predictive classification tasks, especially when the decision boundary can be approximated linearly. It maps high-dimensional input features into a probability space using a weighted linear function followed by a softmax activation. Because it is simple, fast, and highly interpretable, Logistic Regression delivers reliable performance even on complex embeddings like DistilRoBERTa. Its optimization through gradient descent enables it to learn subtle patterns while maintaining stability and minimal overfitting. In this project, Logistic Regression serves as the proposed final model due to its strong accuracy, low computational cost.



FIGIRE 4.1.7 : Internal work flow of Logistic regression : Context-Aware Brand Name classification using DistilBERT Enchanced NLP features from corporate website metadata

Dataset Description

The dataset used in this project contains textual data related to brand names and their corresponding categories. It is stored in a structured format such as CSV or Excel and is used to train and test the machine learning models. Before training, the dataset is cleaned and processed using techniques from Natural Language Processing such as text cleaning, tokenization, and stopword removal.

	A	B	C	D	E	F	G	H
1	SectorId	Sector	IndustryGr	IndustryId	Industry	SubIndustr	SubIndustr	SubIndustryDescription
2	10	Energy	1010	101010	Energy Equ	10101010	Oil & Gas D	Drilling contractors or owners of drilling rigs that contract their services for drill
3	10	Energy	1010	101010	Energy Equ	10101020	Oil & Gas E	Manufacturers of equipment, including drilling rigs and equipment, and provide
4	10	Energy	1010	101020	Oil, Gas &	10102010	Integrated	Integrated oil companies engaged in the exploration & production of oil and gas
5	10	Energy	1010	101020	Oil, Gas &	10102020	Oil & Gas E	Companies engaged in the exploration and production of oil and gas not classif
6	10	Energy	1010	101020	Oil, Gas &	10102030	Oil & Gas F	Companies engaged in the refining and marketing of oil, gas and/or refined proc
7	10	Energy	1010	101020	Oil, Gas &	10102040	Oil & Gas S	Companies engaged in the storage and/or transportation of oil, gas and/or refir
8	10	Energy	1010	101020	Oil, Gas &	10102050	Coal & Cor	Companies primarily involved in the production and mining of coal, related proc
9	15	Materials	1510	151010	Chemicals	15101010	Commodit	Companies that primarily produce industrial chemicals and basic chemicals. I
10	15	Materials	1510	151010	Chemicals	15101020	Diversified	Manufacturers of a diversified range of chemical products not classified in the
11	15	Materials	1510	151010	Chemicals	15101030	Fertilizers	Producers of fertilizers, pesticides, potash or other agriculture-related chemic
12	15	Materials	1510	151010	Chemicals	15101040	Industrial (	Manufacturers of industrial gases.
13	15	Materials	1510	151010	Chemicals	15101050	Specialty (	Companies that primarily produce high value-added chemicals used in the mar
14	15	Materials	1510	151020	Constructi	15102010	Constructi	Manufacturers of construction materials including sand, clay, gypsum, lime, ag
15	15	Materials	1510	151030	Containers	15103010	Metal, Gla	Manufacturers of metal, glass or plastic containers. Includes corks and caps.
16	15	Materials	1510	151030	Containers	15103020	Paper & Pl	Manufacturers of paper and cardboard containers and packaging.
17	15	Materials	1510	151040	Metals & M	15104010	Aluminum	Producers of aluminum and related products, including companies that mine o
18	15	Materials	1510	151040	Metals & M	15104020	Diversified	Companies engaged in the diversified production or extraction of metals and m
19	15	Materials	1510	151040	Metals & M	15104025	Copper	Companies involved primarily in copper ore mining.
20	15	Materials	1510	151040	Metals & M	15104030	Gold	Producers of gold and related products, including companies that mine or proc

Figure 4.1.8: Screenshot of the features in the transaction dataset : Context-Aware Brand Name classification using DistilBERT Enchanced NLP features from corporate website metadata

4.2 SAMPLE CODE

```
import os
import pandas as pd
import numpy as np
import joblib
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
import seaborn as sns
import json
import io
import base64
from flask import Flask, render_template, request, redirect, url_for, flash
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix
# ------------------ Flask App Config ------------------
app = Flask(_name_)
app.secret_key = "supersecretkey"
MODEL_DIR = "model"
RESULTS_DIR = "results"
os.makedirs(MODEL_DIR, exist_ok=True)
os.makedirs(RESULTS_DIR, exist_ok=True)
# ------------------ Helper Functions ------------------
def load_model(model_name):
    model_path = os.path.join(MODEL_DIR, model_name)
    if os.path.exists(model_path):
  return joblib.load(model_path)
    else:
        raise FileNotFoundError(f"Model '{model_name}' not found in {MODEL_DIR}/")
def calculate_metrics(y_true, y_pred):
    return {
        "Accuracy": round(accuracy_score(y_true, y_pred), 3),
        "Precision": round(precision_score(y_true, y_pred, average="weighted", zero_division=0), 3),
        "Recall": round(recall_score(y_true, y_pred, average="weighted", zero_division=0), 3),
        "F1-Score": round(f1_score(y_true, y_pred, average="weighted", zero_division=0), 3)
```

```
    }

def plot_confusion_matrix(y_true, y_pred, labels, title="Confusion Matrix"):
    cm = confusion_matrix(y_true, y_pred)
    plt.figure(figsize=(5,4))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=labels, yticklabels=labels)
    plt.title(title)
    plt.xlabel("Predicted")
    plt.ylabel("Actual")
    plt.tight_layout()
    filename = os.path.join(RESULTS_DIR, f"{title.replace(' ', '_')}.png")
    plt.savefig(filename)
    plt.close()
    return filename
def preprocess_text_data(df):
    # Dummy preprocessing (replace this with your real text cleaning logic)
    if 'text' in df.columns:
        df['clean_text'] = df['text'].str.lower().str.replace(r'[^a-z\s]', '', regex=True)
    else:
        df['clean_text'] = df.iloc[:, 0].astype(str).str.lower()
    return df
# ------------------ Routes ------------------
@app.route('/')
def home():
    return render_template('home.html')
@app.route('/dashboard')
def dashboard():
    plots = [f for f in os.listdir(RESULTS_DIR) if f.endswith(".png")]
    return render_template('dashboard.html', plots=plots)
@app.route('/analysis')
def analysis():
    # Example: load saved metrics (you can expand)
    metrics_path = os.path.join(RESULTS_DIR, "model_metrics.json")
    if os.path.exists(metrics_path):
        with open(metrics_path, "r") as f:
            metrics = json.load(f)
```

```
    else:
        metrics = {"message": "No metrics available yet. Train or test a model first."}
    return render_template("analysis.html", metrics=metrics)
@app.route('/predict', methods=['GET', 'POST'])
def predict():
    if request.method == 'POST':
        file = request.files.get('file')
        if not file:
            flash('Please upload a CSV file!', 'danger')
            return redirect(request.url)
        df = pd.read_csv(file)
        df = preprocess_text_data(df)
        model_name = "Distil_RoBERT-WE_IndustryGroup_LR_model.pkl"  # Adjust to your model
        model = load_model(model_name)
@app.route('/dashboard')
def dashboard():
    plots = [f for f in os.listdir(RESULTS_DIR) if f.endswith(".png")]
    return render_template('dashboard.html', plots=plots)
@app.route('/analysis')
def analysis():
    # Example: load saved metrics (you can expand)
    metrics_path = os.path.join(RESULTS_DIR, "model_metrics.json")
    if os.path.exists(metrics_path):
        with open(metrics_path, "r") as f:
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@app.route('/predict', methods=['GET', 'POST'])
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        if not file:
            flash('Please upload a CSV file!', 'danger')
            return redirect(request.url)
```

```
        df = pd.read_csv(file)
        df = preprocess_text_data(df)
        model_name = "Distil_RoBERT-WE_IndustryGroup_LR_model.pkl"  # Adjust to your model
        model = load_model(model_name)
        # Example placeholder encoding
        features = df['clean_text'].apply(lambda x: len(x.split())).values.reshape(-1, 1)
        y_pred = model.predict(features)
        df['Predicted_Label'] = y_pred
        # Example metrics (if you have ground truth)
        if 'label' in df.columns:
            metrics = calculate_metrics(df['label'], y_pred)
            with open(os.path.join(RESULTS_DIR, "model_metrics.json"), "w") as f:
                json.dump(metrics, f, indent=4)
            plot_confusion_matrix(df['label'], y_pred, labels=np.unique(df['label']))
        else:
            metrics = {"Info": "No true labels found for metrics."}
        html_table = df.head(20).to_html(classes='table table-striped', index=False)
        return render_template('predict.html', tables=[html_table], metrics=metrics)
    return render_template('predict.html')
        if 'label' in df.columns:
            metrics = calculate_metrics(df['label'], y_pred)
            with open(os.path.join(RESULTS_DIR, "model_metrics.json"), "w") as f:
                json.dump(metrics, f, indent=4)
            plot_confusion_matrix(df['label'], y_pred, labels=np.unique(df['label']))
        else:
            metrics = {"Info": "No true labels found for metrics."}
        html_table = df.head(20).to_html(classes='table table-striped', index=False)
        return render_template('predict.html', tables=[html_table], metrics=metrics)
    return render_template('predict.html')
if __name___== '__main__':
    app.run(debug=True)
```

5. RESULTS & DISCUSSION

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The proposed system successfully analyzes and evaluates brand names using Natural Language Processing and machine learning techniques. Text preprocessing methods such as tokenization, stopword removal, and lemmatization are applied to clean the data. The DistilRoBERTa model extracts meaningful features from the text, which are then used by machine learning algorithms like Logistic Regression, Random Forest, and SVM for classification. The system produces accurate predictions and helps users evaluate brand names efficiently. The Flask-based web interface allows users to input brand names and obtain results in real time.

Text preprocessing methods such as tokenization, stopword removal, and lemmatization are applied to clean the data. The DistilRoBERTa model extracts meaningful features from the text, which are then used by machine learning algorithms like Logistic Regression, Random Forest, and SVM for classification. The system produces accurate predictions and helps users evaluate brand names efficiently. Natural Language Processing and machine learning techniques. Processing and machine learning techniques. Text preprocessing methods such as tokenization, stopword removal, and lemmatization are applied to clean the data. The DistilRoBERTa model extracts meaningful features from the text.

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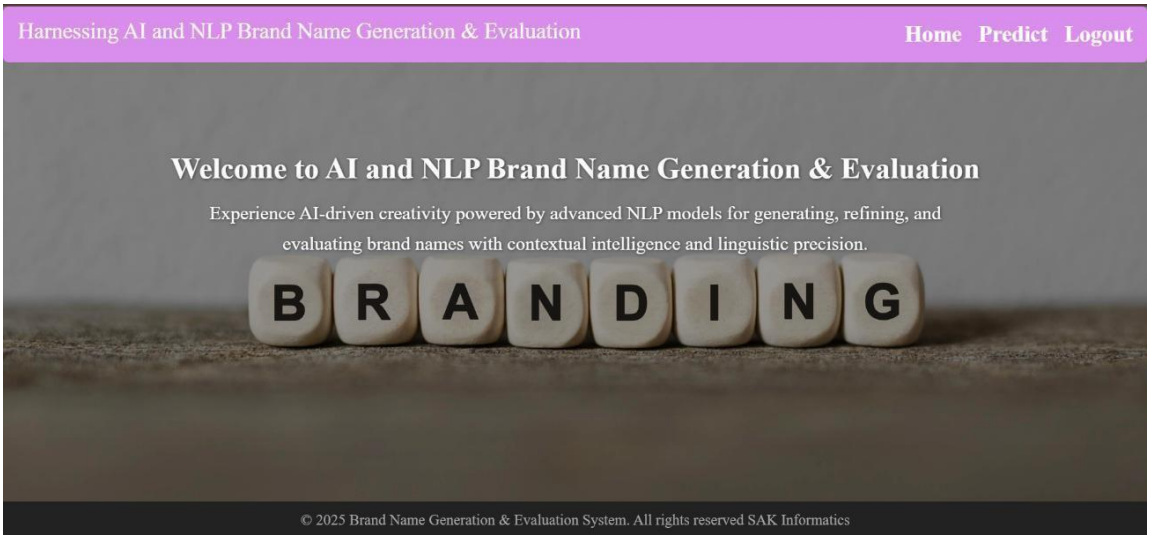
5.1 System Home Page:

shows the user registration interface, where essential details such as username, email, contact information, address, and password are collected to create a new account. The structured arrangement of input areas ensures orderly data entry and minimizes ambiguity during form completion. This interface supports a streamlined onboarding flow, preparing users for authenticated access to system features. Its clean alignment of fields enhances usability and reduces cognitive load. The form establishes the preliminary data requirements needed for personalized interaction within the platform.

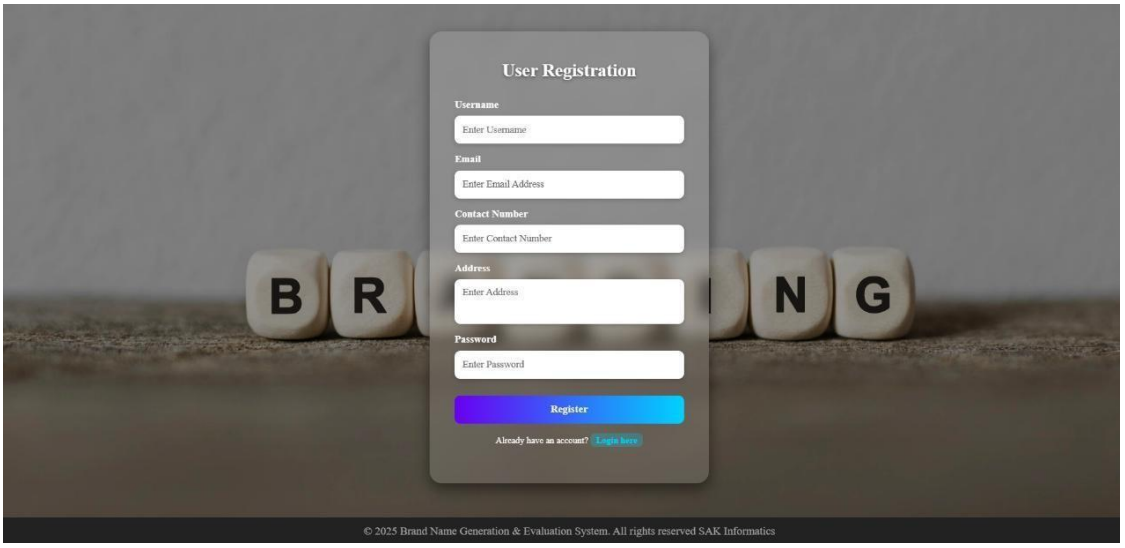


Screenshot 5.1 : System Home page of Context-Aware Brand Name Classification using DistilBERT-Enhanced NLP Features from corporate Websites Metadata

The Home Page is the initial interface of the system where users are introduced to the Brand Name Generation and Evaluation System. It provides a brief overview of the project, its purpose, and features. The page includes navigation options such as registration, login, dataset upload, and prediction modules. It is designed to be simple and user-friendly, allowing users to easily access different functionalities of the system. The home page is developed using the Flask framework, ensuring smooth interaction between the frontend and backend components.

5.2 Registration Page:

shows the user registration interface, where essential details such as username, email, contact information, address, and password are collected to create a new account. The structured arrangement of input areas ensures orderly data entry and minimizes ambiguity during form completion. This interface supports a streamlined onboarding flow, preparing users for authenticated access to system featurees.

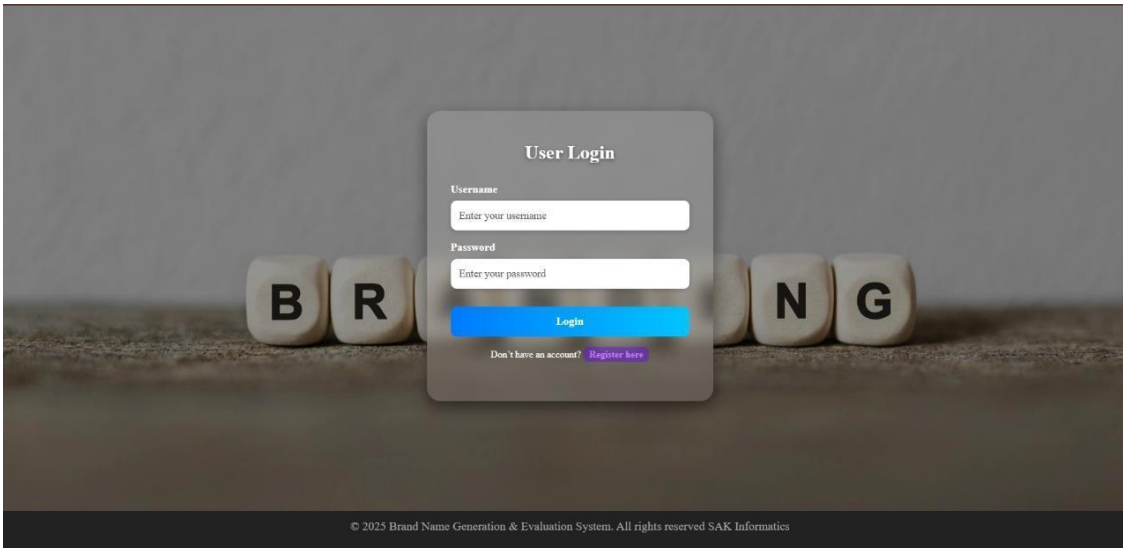


Screenshot 5.2 : Registration Page of Context-Aware Brand Name Classification using DistilBERT-Enhanced NLP Features from corporate Websites Metadata

The Registration Page allows new users to create an account in the system. Users are required to enter details such as username, email, and password. The system validates the entered information and stores it securely in the database. This page ensures that only authorized users can access the system features. It plays an important role in maintaining data privacy and system security by restricting access to registered users only.

5.3 Login Page

shows the login interface through which authenticated users gain access to the system's internal modules. The minimal two-field structure emphasizes direct credential entry, reducing friction and enabling quick account verification. This layout ensures a focused interaction without unnecessary elements, enhancing both accuracy and speed during login. The interface preserves consistency with the registration design, maintaining a unified visual identity. It forms a critical checkpoint for securing user-specific operations within the platform.



Screenshot 5.3 : Login Page of Context-Aware Brand Name Classification using DistilBERT-Enhanced NLP Features from corporate Websites Metadata

The Login Page is used by registered users to access the system. Users enter their credentials such as username and password, which are verified by the system. Once authenticated, users are redirected to the main dashboard or prediction page. This page ensures secure access control and prevents unauthorized usage of the system.

5.4 Uploading Test Dataset Page:

shows a word cloud of the 100 most frequent terms extracted from the processed brand-description corpus, where font size indicates term frequency. Dominant tokens such as "service," "company," "classified," "product," and "consumer" appear prominently, reflecting the business-centric nature of the dataset. The spread of terms highlights recurring patterns related to industries, operational activities, and commercial attributes. The visualization provides a quick qualitative overview of lexical trends before deeper NLP modeling. It effectively validates the thematic structure of the corpus prior to feature extraction.

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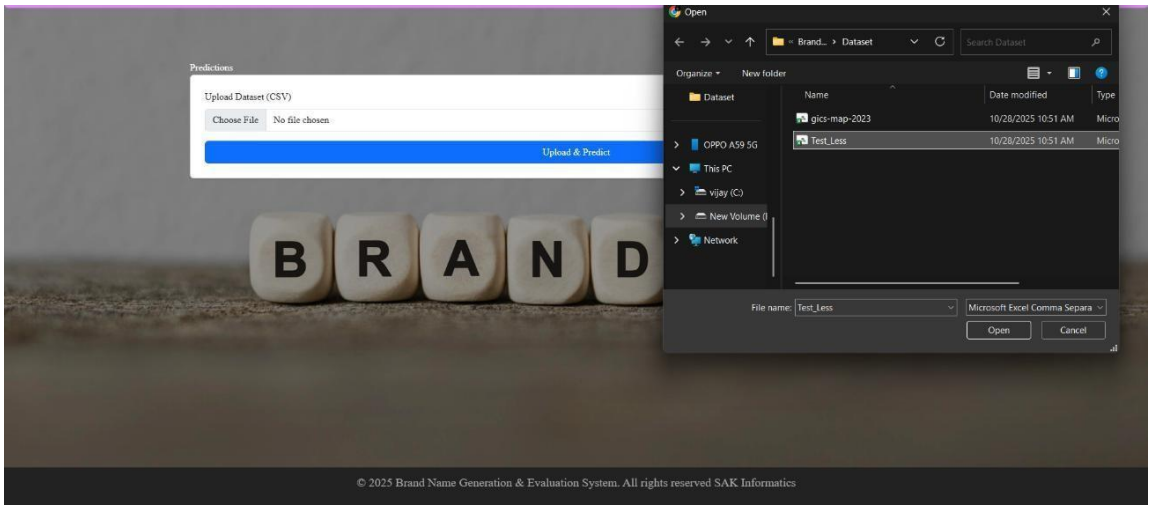
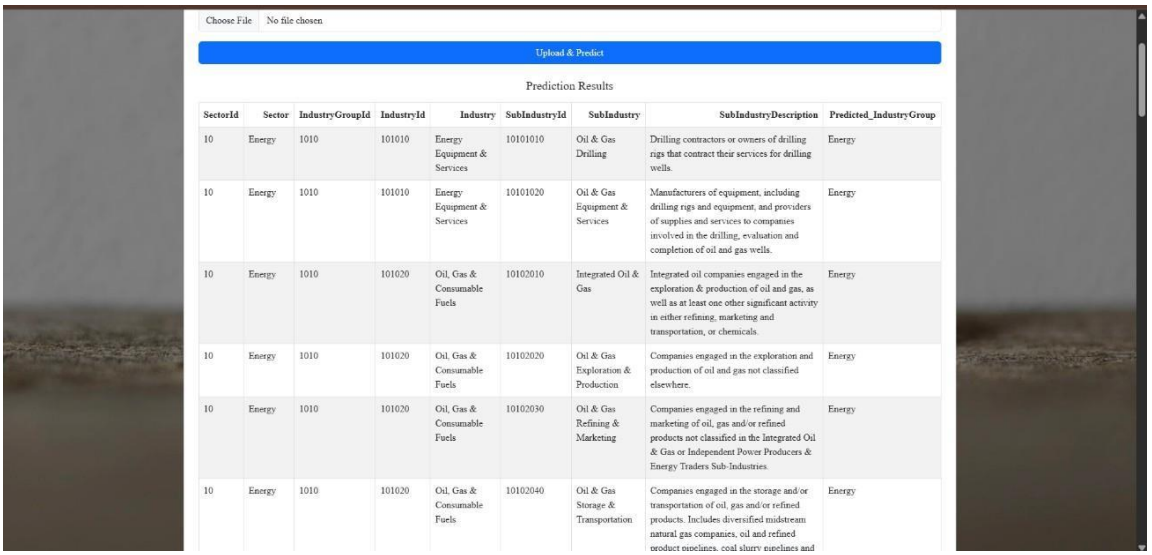


Screenshot 5.4 : Uploading test dataset page of Context-Aware Brand Name Classification using DistilBERT-Enhanced NLP Features from corporate Websites Metadata

The Uploading Test Dataset Page allows users to upload datasets in formats such as CSV or Excel. The uploaded dataset contains brand names or textual data that needs to be analyzed. The system processes the uploaded data using techniques from Natural Language Processing, including preprocessing and feature extraction. The page provides an easy interface for selecting and submitting files, making the process simple for users without technical knowledge.

5.5 Final output:

shows a bar chart representing the 20 most frequent unigrams identified during corpus analysis, arranged in descending order of occurrence. High-frequency terms related to business operations and industry classification stand out clearly, providing structured quantitative insight into vocabulary distribution. This visualization offers a more precise interpretation than the word cloud, supporting comparison of lexical prominence across terms. The chart confirms the consistency of the dataset's thematic focus and strengthens the basis for downstream machine-learning tasks. It serves as an essential validation step for understanding term importance before semantic model.



SectorId	Sector	IndustryGroupId	IndustryId	Industry	SubIndustryId	SubIndustry	SubIndustryDescription	Predicted_IndustryGroup
10	Energy	1010	101010	Energy Equipment & Services	10101010	Oil & Gas Drilling	Drilling contractors or owners of drilling rigs that contract their services for drilling wells.	Energy
10	Energy	1010	101010	Energy Equipment & Services	10101020	Oil & Gas Equipment & Services	Manufacturers of equipment, including drilling rigs and equipment, and providers of supplies and services to companies involved in the drilling, evaluation and completion of oil and gas wells.	Energy
10	Energy	1010	101020	Oil, Gas & Consumable Fuels	10102010	Integrated Oil & Gas	Integrated oil companies engaged in the exploration & production of oil and gas, as well as at least one other significant activity in either refining, marketing and transportation, or chemicals.	Energy
10	Energy	1010	101020	Oil, Gas & Consumable Fuels	10102020	Oil & Gas Exploration & Production	Companies engaged in the exploration and production of oil and gas not classified elsewhere.	Energy
10	Energy	1010	101020	Oil, Gas & Consumable Fuels	10102030	Oil & Gas Refining & Marketing	Companies engaged in the refining and marketing of oil, gas and/or refined products not classified in the Integrated Oil & Gas or Independent Power Producers & Energy Traders Sub-Industries.	Energy
10	Energy	1010	101020	Oil, Gas & Consumable Fuels	10102040	Oil & Gas Storage & Transportation	Companies engaged in the storage and/or transportation of oil, gas and/or refined products. Includes diversified midstream natural gas companies, oil and refined product pipelines, coal slurry pipelines and	Energy

Screenshot 5.5 : Final output of Context-Aware Brand Name Classification using DistilBERT-Enhanced NLP Features from corporate Websites

The Final Output Page displays the results generated by the system after processing the input data. The system uses machine learning models along with embeddings from DistilRoBERTa to analyze the input and produce predictions. The output includes classification results, prediction scores, and evaluation metrics such as accuracy and precision. The results are presented in a clear and understandable format, allowing users to easily interpret the performance of the model and the quality of the brand names.

6. VALIDATION

6. VALIDATION

The validation of the proposed system is performed to ensure that the brand name generation and evaluation model works accurately and efficiently. The dataset used in the project is divided into training and testing sets to evaluate the performance of the machine learning models. After training the models with the extracted features from the DistilRoBERTa transformer model, the system is tested using unseen data to check how well it predicts and evaluates brand names.
Different machine learning algorithms such as Logistic Regression, Random Forest, and Support Vector Machine are applied, and their performances are compared using evaluation metrics like accuracy, precision, recall, and F1-score. These metrics help measure how correctly the system classifies and evaluates brand names based on the trained dataset.

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6.1 INTRODUCTION

Validation is an important step in evaluating the performance and reliability of the developed system. It ensures that the machine learning models used in the project produce accurate and consistent results when applied to unseen data. In this project, validation is performed by testing the trained models on a separate testing dataset and analyzing their performance using different evaluation metrics.

In this project, validation is carried out using a separate test dataset that was not used during the training phase. This helps in avoiding overfitting and provides a realistic measure of the model's performance. The dataset is split into training and testing sets, where the models are trained on the training data and evaluated on the test data.

To validate the performance of the system, multiple evaluation metrics are used. These include accuracy, precision, recall, and F1-score, which provide a comprehensive understanding of the model's classification capability. Accuracy measures the overall correctness of predictions, while precision and recall evaluate the model's ability to correctly identify industry categories without false positives and false negatives. The F1-score balances both precision and recall, making it a reliable metric for performance assessment.

In addition to numerical metrics, confusion matrices are used to visualize the classification performance across different industry groups. The confusion matrix clearly shows how many instances are correctly classified and where misclassifications occur. This helps in identifying specific categories where the model needs improvement.

Overall, the validation results demonstrate that the proposed system effectively classifies brand names into their respective industry categories with high accuracy. The use of advanced NLP techniques such as DistilRoBERT embeddings, combined with machine learning algorithms, ensures robust and reliable predictions. This validation confirms that the system is suitable for real-world applications such as business analytics, market research, and automated classification systems.

6.2 TEST CASES

TABLE 6.2.1 UPLOADING TRANSACTION DATASET

Test case ID	Test case name	Purpose	Test Case	Output
1	Upload Dataset	To verify that the system successfully uploads the dataset for processing	User uploads dataset file in CSV format	Dataset is successfully uploaded and ready for preprocessing (Pass)

TABLE 6.2.2 CLASSIFICATION

Test Case ID	Test Case Name	Purpose	Input	Output
1	User Registration (Valid)	To verify that a new user can successfully register in the system	User enters valid username, email, password and details	User account is successfully created (Pass)
2	User Registration (Duplicate Username)	To check whether the system prevents registration with an existing username	User enters an already registered username with other details	System displays message "Username already exists" and registration fails **(Fail)**
3	User Login (Valid)	To verify that registered users can log in to the system	Registered username and correct password	User successfully logs into the system dashboard **(Pass)**
4	User Login (Invalid)	To verify that the system rejects incorrect login credentials	Registered username with wrong password	System displays Invalid login details and access is denied **(Fail)**
5	Text Data Preprocessing	To verify that the system cleans and processes input text using NLP techniques	Dataset containing raw text data	System performs tokenization, stopword removal and lemmatization successfully (Pass)

6	Feature Extraction	To verify that the system generates embeddings using DistilRoBERTa	Preprocessed text data	System generates semantic embeddings for classification (Pass)
7	Model Training	To verify that machine learning models are trained correctly	DistilRoBERTa feature vectors	Models such as Logistic Regression, Random Forest and SVM are trained successfully (Pass)
8	Model Evaluation	To verify system calculates performance metrics	Trained model predictions	Accuracy, Precision, Recall and F1-Score are generated (Pass)
9	Prediction Generation	To verify the system predicts industry group for new input data	User uploads new dataset	System displays predicted industry group results (Pass)
10	Result Visualization	To verify that the system displays results through visualizations	Model evaluation metrics	System shows graphs like word cloud, bigrams and confusion matrix (Pass)

7. CONCLUSION & FUTURE ASPECTS

7. CONCLUSION & FUTURE ASPECTS

The project demonstrates the use of Artificial Intelligence and Natural Language Processing techniques for analyzing and evaluating brand names. By applying preprocessing techniques such as tokenization, stopword removal, and lemmatization, the system prepares textual data for analysis. The DistilRoBERTa model is used for feature extraction to capture semantic relationships in text, while machine learning algorithms like Logistic Regression, Random Forest, and Support Vector Machine perform classification and prediction. The results show that the system can effectively analyze brand names and provide reliable predictions. The integration of the model with a Flask-based web interface allows users to easily interact with the system and obtain results in real time.

7.1 PROJECT CONCLUSION

The proposed AI-based Brand Name Generation and Evaluation System provides an efficient and automated solution for analyzing brand names. The system combines NLP techniques, transformer-based feature extraction, and machine learning algorithms to improve the accuracy and effectiveness of brand name evaluation. The implementation of data analysis techniques and visualization methods helps understand patterns in brand names. The developed web application makes the system user-friendly and practical for real-world applications. Overall, the project demonstrates how AI and NLP technologies can assist businesses in selecting meaningful and effective brand names.

This project presents an intelligent system for brand name generation and evaluation using techniques from Artificial Intelligence and Natural Language Processing. The system processes textual data through preprocessing steps and extracts meaningful features using transformer-based embeddings. These features help machine learning algorithms understand the semantic relationships between words and improve classification accuracy.

Advanced language models such as DistilRoBERTa are used to generate contextual embeddings from text, which are then analyzed using machine learning algorithms like Logistic Regression, Support Vector Machine, and Random Forest. The entire system is integrated with a web interface using Flask, allowing users to upload data and obtain predictions easily. The experimental results demonstrate that the proposed system can effectively analyze textual data and assist in evaluating.

7.2 FUTURE ASPECTS

The system can be expanded by integrating larger transformer models such as RoBERTa-large or domain-adapted financial/industry-specific LLMs for improved semantic understanding. The dataset itself can be broadened to include more sectors, multilingual descriptions, and real-world organizational profiles to improve generalization. Future versions may incorporate hierarchical multi-level classification so the model predicts Sector, IndustryGroup, Industry, and SubIndustry simultaneously. Deployment can be scaled using containerized microservices on cloud platforms to support enterprise-grade throughput. The system can also evolve into an automated analytics engine that provides insights, similarity matching, and industry trend detection using advanced LLMreasoning.

The brand name generation system by integrating more advanced techniques from **Artificial Intelligence** and **Natural Language Processing**. In the future, the system can be improved by incorporating larger transformer models and deep learning architectures to generate more creative and context-aware brand name suggestions. The application can also be extended to support multiple languages, allowing businesses from different regions to generate suitable brand names. Additionally, integrating the system with real-time market analysis and domain name availability checking can help users select unique and commercially viable brand names. The project can also be deployed as a scalable cloud-based platform with enhanced user interfaces, making it accessible to startups, entrepreneurs, and marketing professionals worldwide.

The proposed system can be further improved by integrating more advanced techniques from Artificial Intelligence and Natural Language Processing to enhance the accuracy and efficiency of brand name evaluation. Future work can focus on experimenting with more powerful transformer-based models and larger datasets to capture deeper semantic relationships in textual data. This can help the system generate more creative, relevant, and meaningful brand name suggestions for different industries.

Another possible improvement is the development of a more advanced user interface using modern web technologies along with Flask to provide better visualization and interactive features. The system can also be integrated with external databases or online platforms to analyze trending brand names and market preferences.

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8.2 GITHUB LINK

https://github.com/PrachetanaGodishala/NLP-Brand-Name-Industry-Classification